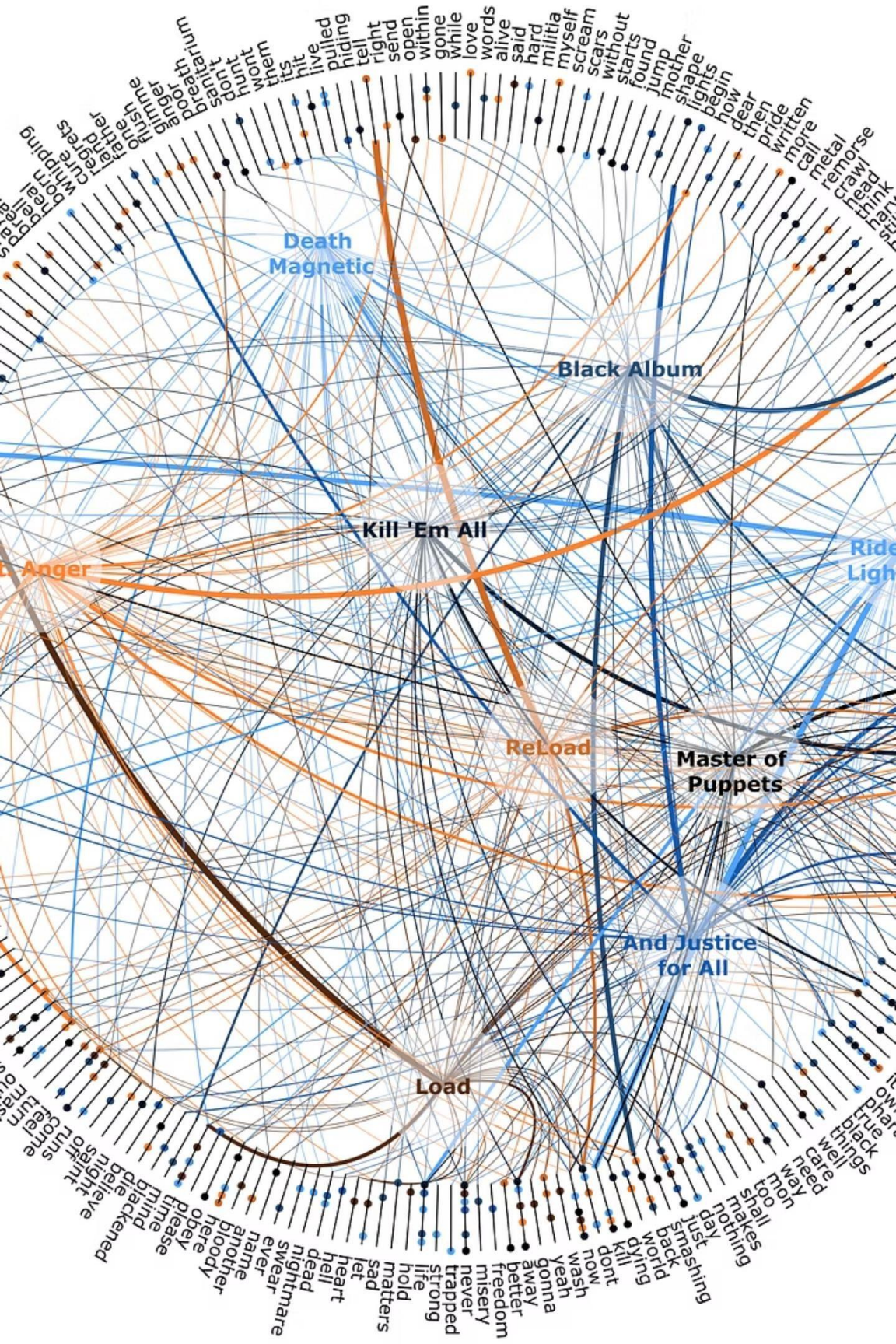




# Music Listener Segmentation using Streaming Patterns with Time-Based Analysis and Visualization

Presenters:Sanjana and madhu shree

*Speaker Notes: Welcome the audience and introduce yourselves. Set the context by mentioning this is a data science project analysing real streaming behaviour.*



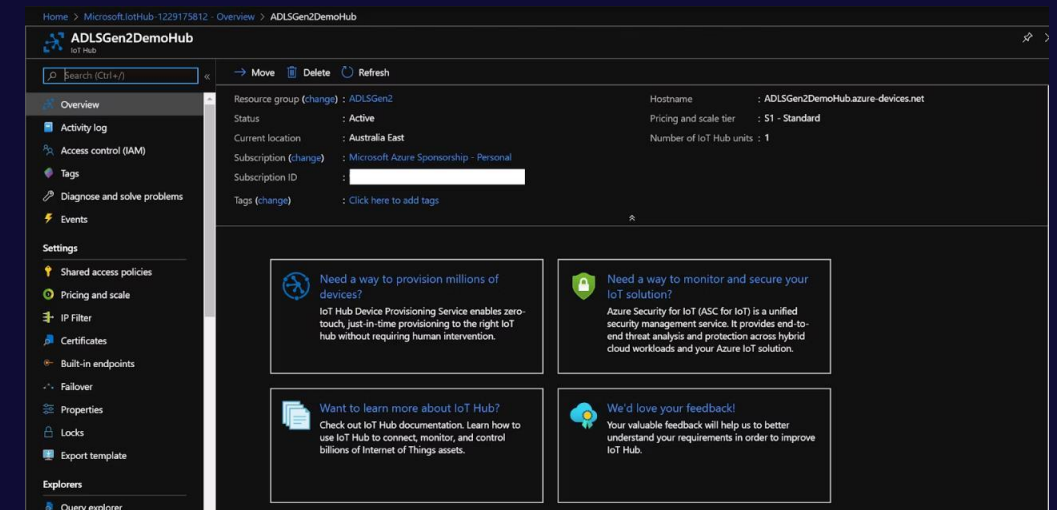
# Project Overview: Understanding Music Listening Behaviour

Music streaming platforms generate vast amounts of data on user behaviour every second. This project aims to segment music listeners based on their streaming patterns, with a special focus on time-based listening habits. By analysing when users listen, how long they engage, and their skipping patterns, we can identify distinct listener personas.

## Why This Matters

- Personalised recommendations improve user retention
- Understanding temporal patterns helps optimise platform features
- Segmentation enables targeted marketing strategies
- Insights can guide content acquisition decisions

*Speaker Notes: Explain the business value of user segmentation. Mention how streaming platforms like Spotify and Gaana use similar techniques for personalisation.*



# Dataset & Features

Our dataset captures streaming events with rich behavioural information. Each record represents a single track play event, providing granular insights into listening patterns.

## Core Identifiers

- id: Unique user identifier
- timestamp: Exact time of play
- track\_name: Song title
- artist: Music artist name

## Behavioural Metrics

- duration\_ms: Track length in milliseconds
- skipped: Binary flag (1=skipped, 0=fully listened)
- platform: Streaming platform used

## Derived Features

- hour: Extracted from timestamp
- time\_segment: Morning/Afternoon/Evening/Night
- listening\_frequency: Plays per time period
- skip\_rate: Proportion of skipped tracks

*Speaker Notes: Walk through each feature and explain its relevance. Emphasise that timestamp and skip behaviour are particularly valuable for analysis.*

# Data Preprocessing Pipeline

1

### Missing Values

Identified and handled missing data points. Duration missingness addressed using median imputation for robustness.

2

### Timestamp Conversion

Converted string timestamps to datetime format for temporal analysis and extraction of time components.

3

### Feature Extraction

Extracted hour component from timestamp for time-based segmentation and analysis.

## Key Preprocessing Steps

1. Data quality assessment and missing value identification
2. Duration median calculation and imputation
3. String-to-datetime conversion using pandas
4. Hour extraction using datetime.hour attribute

## Quality Checks Applied

- Null value verification
- Data type validation
- Outlier detection
- Consistency verification

Speaker Notes: Explain why median imputation was chosen over mean (robustness to outliers). Mention pandas library for datetime operations.

+

Search

for working with messy data.

OverConfigure Parsing OptionsProject nameOpenRefine sample csv

| rs_num | Sample_Des | Last_name | First_name | Street_birth | Age | Date_admitted | Date_discharge     | Committing_party                            | Occupation |
|--------|------------|-----------|------------|--------------|-----|---------------|--------------------|---------------------------------------------|------------|
|        | A          | Valaly    | Thomas     | duane        | 21  | 1/29/1846     | 1/29/1846 0:00:00  | Comm per Agent                              | Tailor     |
|        | A          | Long      | John       | duane        | 23  | 2/27/1846     | 2/27/1846 0:00:00  | Comm per Agent                              | Laborer    |
|        | A          | Madden    | Ellen      | Forrest      | 22  | 2/5/1846      | 2/5/1846 0:00:00   | Superintendent per Alderman Henry, 6th ward | Married    |
|        | A          | Mitchell  | Fanny      | simpson      | 23  | 4/8/1846      | 4/8/1846 0:00:00   | Comm per Agent                              | Married    |
|        | A          | Gormley   | Hugh       | simpson      | 22  | 8/31/1846     | 8/31/1846 0:00:00  | Comm per Agent                              | Laborer    |
|        | A          | Murray    | Eliza      | forrest      | 30  | 4/14/1846     | 4/14/1846 0:00:00  | Comm per Agent                              | Widow      |
|        | A          | Smithwick | George     | Simpson      | 62  | 6/29/1846     | 6/29/1846 0:00:00  | Superintendent per Alderman Jackson         | Laborer    |
|        | A          | Collins   | Patrick    | first        | 23  | 8/14/1846     | 8/14/1846 0:00:00  | Anderson                                    | Laborer    |
|        | A          | Gilligan  | Mary       | First        | 35  | 4/10/1847     | 4/10/1847 0:00:00  | Anderson per Witherell                      | Spinster   |
|        | A          | Leonard   | John       | first        | 20  | 3/25/1847     | 3/25/1847 0:00:00  | Anderson                                    | Laborer    |
|        | A          | Moran     | Michael    | Mercer       | 25  | 11/18/1846    | 11/18/1846 0:00:00 | Anderson                                    | Laborer    |
|        | A          | Ward      | Michael    | Mercer       | 11  | 4/13/1847     | 4/13/1847 0:00:00  | Superintendent per Alderman Smith, 7th ward | 999        |
|        | A          | Monaghan  | Michael    | third        | 20  | 7/12/1847     | 7/12/1847 0:00:00  | Anderson                                    | Laborer    |
|        | A          | Delaney   | John       | Third        | 8   | 3/2/1847      | 3/2/1847 0:00:00   | Comm per Agent                              | 999        |
|        | A          | Clancy    | Timothy    | Mercer       | 27  | 3/25/1847     | 3/25/1847 0:00:00  | Comm per Agent                              | Carpenter  |
|        | A          | Stevens   | Martin     | first        | 50  | 1/25/1847     | 1/25/1847 0:00:00  | Comm per Agent                              | Laborer    |
|        | A          | Kelly     | Arthur     | Duane        | 25  | 1/29/1847     | 1/29/1847 0:00:00  | Comm per Agent                              | Laborer    |
|        | A          | Rafferty  | Barnard    | Dauane       | 51  | 5/31/1847     | 5/31/1847 0:00:00  | Anderson per Witherell                      | Mason      |
|        | A          | Burke     | John       | third        | 26  | 5/21/1847     | 5/21/1847 0:00:00  | Anderson per                                | Laborer    |

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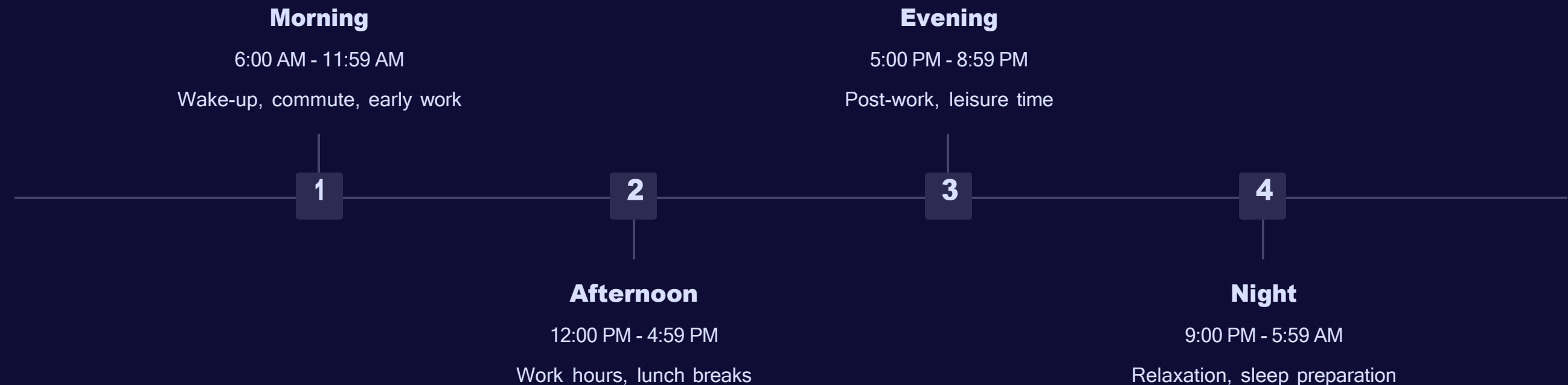
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# Time-Based Analysis: A Unique Feature

One of the key innovations in this project is the temporal segmentation of listening behaviour. We categorised the 24-hour day into four distinct segments based on typical human activity patterns.



## Insight from Analysis

Preliminary histogram analysis revealed peak listening hours between 6 PM and 10 PM, with a secondary peak during morning commute (7 AM - 9 AM). This temporal pattern informed our segmentation strategy and revealed distinct listener types based on time preferences.

*Speaker Notes: Emphasise that time-based analysis is often overlooked but provides valuable context. Mention the histogram showed clear daily patterns that could be leveraged for personalisation.*

# Feature Engineering for Clustering

To enable effective user segmentation, we engineered several derived features that capture different aspects of listening behaviour. These features form the basis for clustering analysis.



## Listening Frequency

Number of tracks played per time period, capturing engagement intensity. Calculated for each user across all time segments to identify peak activity windows.



## Average Duration

Mean listening time per track, indicating depth of engagement. Helps distinguish between casual browsing and focused listening sessions.



## Skip Rate

Proportion of tracks skipped versus completed, revealing satisfaction and interest levels. Critical metric for understanding content relevance.



## Time-Based Behaviour

Activity distribution across morning, afternoon, evening, and night segments. Reveals lifestyle patterns and routine listening habits.

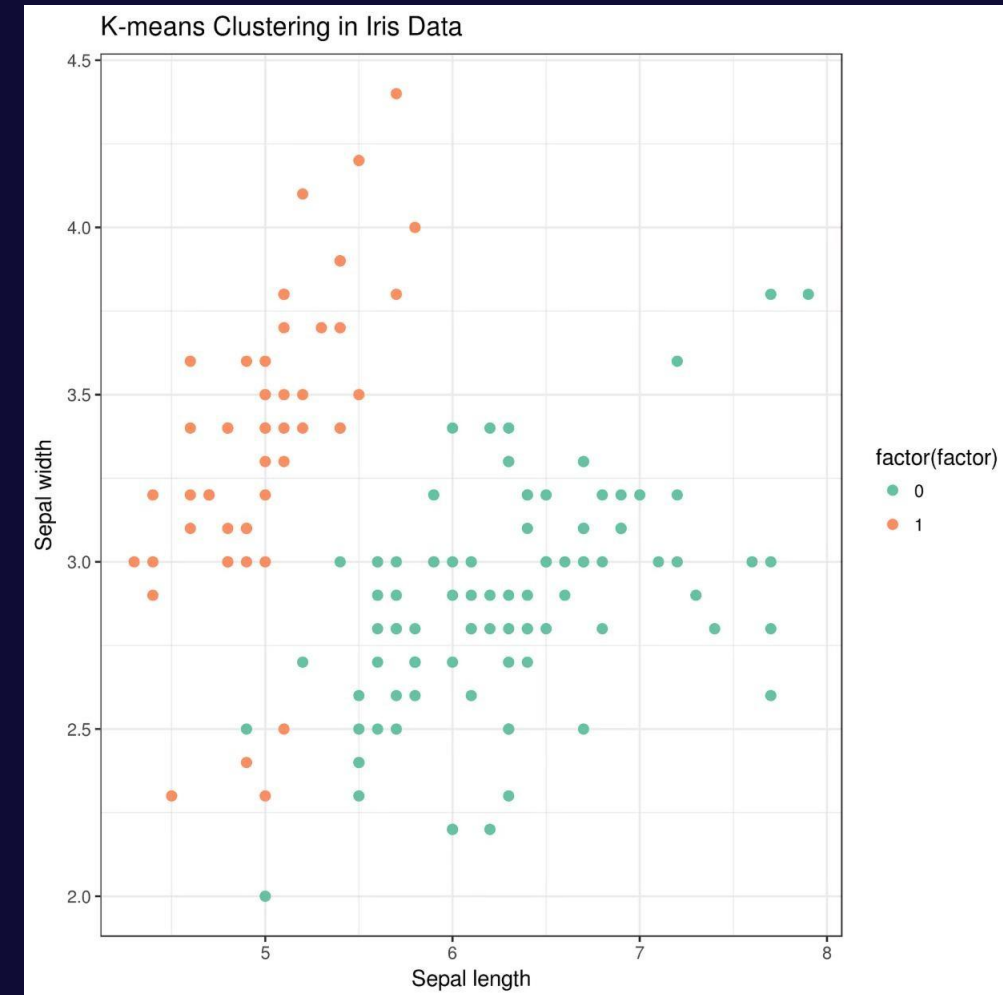
*Speaker Notes: Explain that these features were normalised before clustering. Skip rate is particularly important as it directly indicates user satisfaction.*

# K-Means Clustering: The Algorithm Explained

K-Means is an unsupervised learning algorithm that partitions data into K distinct clusters based on feature similarity. Think of it as grouping similar items together without knowing the groups beforehand.

## How It Works (Simplified)

1. Choose K: Decide how many clusters you want (we used elbow method)
2. Place centroids: Randomly position cluster centres in feature space
3. Assign points: Each user assigned to nearest centroid
4. Update centroids: Move centres to average position of assigned users
5. Repeat: Steps 3-4 until centroids stabilise



## Key Considerations

- Features must be normalised
- Elbow method helps choose K
- Algorithm can converge to local minima
- Results depend on initial centroid placement

*Speaker Notes: Use the balloon analogy - imagine blowing up balloons until they touch; the centres are your clusters. Mention we used StandardScaler for normalisation.*

# PCA: Visualising High-Dimensional Data

Principal Component Analysis (PCA) helps us visualise clustering results by reducing our high-dimensional feature space into 2-3 dimensions while preserving most of the variance.

01

## Standardise Data

Apply StandardScaler to ensure all features contribute equally to PCA transformation

03

## Transform Features

Project original features onto principal component axes

## Benefits of PCA

- Reduces noise and redundancy
- Enables 2D/3D visualisation
- Speeds up clustering
- Reveals data structure

*Speaker Notes: Explain that PCA helps validate clustering by showing clear separation. Mention the explained variance ratio helps choose number of components.*

02

## Compute Components

Find principal components that capture maximum variance in the data

04

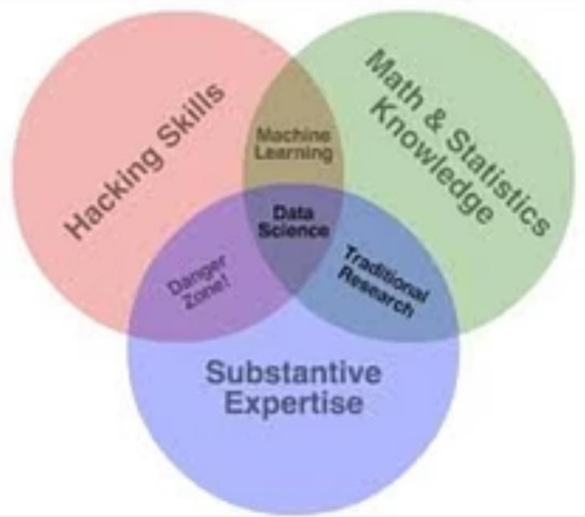
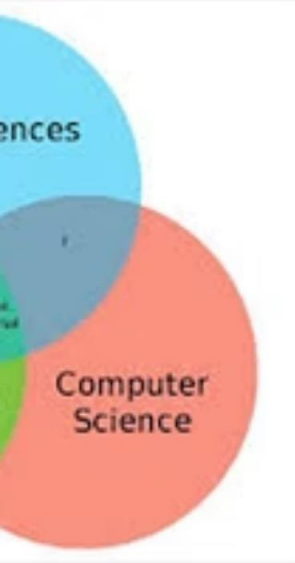
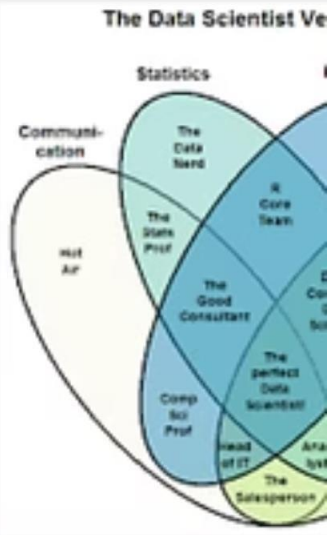
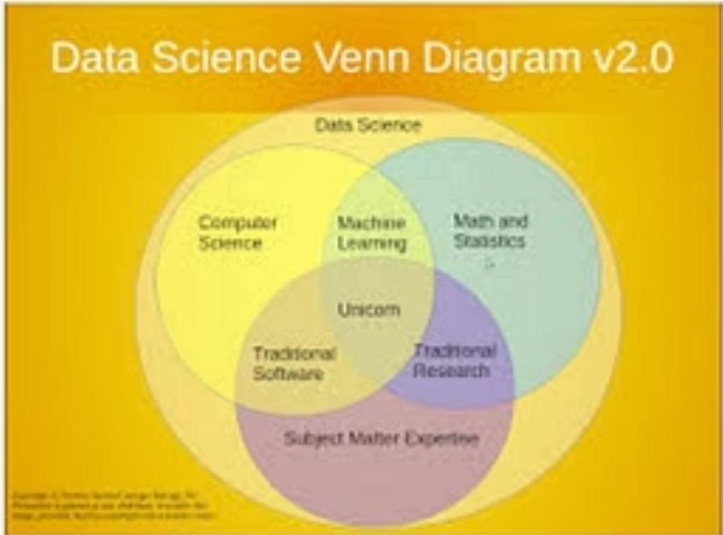
## Visualise Clusters

Plot users in 2D space coloured by cluster assignment

## Implementation Details

- Used scikit-learn PCA
- Retained components explaining ~85% variance
- Standardised before applying PCA





# Results & Insights: Identifying Listener Personas

Our clustering analysis revealed four distinct listener segments with unique behavioural patterns. These insights can directly inform platform personalisation strategies.



## Heavy Engagers

- High frequency across all time periods
- Long average listening duration
- Low skip rate (7-10%)
- Evening-focused activity



## Casual Listeners

- Moderate frequency, random timing
- Average duration engagement
- Medium skip rate (15-20%)
- No clear time preference



## High-Skip Users

- Very high skip rate (25-35%)
- Short listening duration
- Exploratory browsing behaviour
- Morning and afternoon peaks



## Night Owls

- Concentrated late-night activity
- Moderate-to-high frequency
- Varied skip patterns
- Unique time segment preference

Speaker Notes: Connect these personas to business applications. Heavy engagers are retention priority, high-skip users need better recommendations, night owls need specialised content.



# Conclusion & Future Directions

## Key Takeaways

- Time-based analysis reveals distinct listening patterns missed by traditional methods
- Four listener personas identified with clear behavioural differences
- Clustering + PCA enables effective visualisation and interpretation
- Insights directly applicable to platform personalisation

## Project Achievements

- Complete data preprocessing pipeline
- Novel time-based feature engineering
- Validated clustering approach
- Interpretable results with business value

## Future Enhancements

Incorporate audio features (genre, tempo, energy) and demographic data for richer segmentation. Explore hierarchical clustering for multi-level user taxonomy.

## Business Applications

Deploy segments for personalised playlists, targeted notifications, and content acquisition. Monitor segment evolution over time for trend detection.

*Speaker Notes: Thank the audience. Open for questions about methodology, results, or implementation details. Mention dataset limitations and scope for extension.*